

detected motion, the motion frame-binary image, and also the regions where motion is detected. Motion processing algorithms are designed to perform post-processing of detected motion—to highlight motion regions, count moving objects and track objects.

Motion detection algorithms: The AForge.NET library comprises various motion detection algorithms like the two-frame difference motion detector, the simple background modelling motion detector, and the custom frame difference motion detector.

Motion processing algorithms: The AForge.NET library consists of various motion processing algorithms—motion area highlighting, motion border highlighting, grid motion area processing, and blob counting objects processing.

AForge.Math

This library comprises various mathematics related algorithms like:

- 3D pose estimation using POSIT and Coplanar POSIT algorithms
- Simple shape checker
- Graham convex hull

AForge.Video

This library comprises various classes to process all sorts of video data. It has the following components:

- Access to JPEG and MJPEG streams - IP cameras
- Access to USB Web cameras - captures video from the DirectShow Interface
- Reading/writing of AVI files
- Reading/writing video files using the FFmpeg library
- Microsoft Kinect sensor support
- XIMEA camera support

AForge.Robotics

The following robotics kits/boards are supported by this library:

- Qwerk robotics board
- Lego Mindstorm RCX robotics kit
- Lego Mindstorm NXT robotics kit
- Surveyor SRV-1 Blackfin robot
- Surveyor Stereo Vision Systems robotics board

AForge.Neuro

This library consists of neural network architectures along with varied learning algorithms, such as:

- Multi-layer feed forward networks
- Distance networks
- Simple perceptron's learning, Delta rule learning, back propagation learning, Kohonen SOM learning, evolutionary learning based on genetic algorithms
- Threshold, Sigmoid, Bipolar Sigmoid based activation functions

AForge.Genetic

This library makes use of various genetic algorithms, genetic programming and gene expression areas like:

- GA chromosomes (binary, short array and double array), GP tree-based chromosome and GEP chromosome
- Selection algorithms (elite, roulette wheel and rank)
- Common fitness functions (1/2D function optimisation, symbolic regression and time series prediction)
- Population class to handle chromosomes

AForge.Fuzzy

This library has been primarily designed to perform fuzzy computations by using basic fuzzy sets and linguistic variables with a complete inference system.

AForge.MachineLearning

This library supports various machine learning based algorithms like:

- QLearning and Sarsa learning
- Epsilon Greedy, Boltzmann, Roulette wheel and Tabu Search exploration

The AForge.NET framework currently consists of the following two tools:

- Image processing prototypers
- Debugging visualisers

Image processing prototypers

These were primarily designed to process image processing and computer vision algorithms faster, focusing on algorithms only. They provide a generic user interface to help developers test algorithms on image data sets and inspect the results.

Debugging visualisers

The AForge.NET framework consists of a few visualiser components, which are used for debugging image processing and computer vision algorithms. The following are the visualisers available currently:

- System.Drawing.Image
- System.Drawing.Imaging.BitmapData
- AForge.Imaging.UnmanagedImage
- AForge.Math.Histogram 

References

- [1] <http://www.aforgenet.com>
- [2] <https://github.com/andrewkirillov/AForge.NET>

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